Activity 5 – Database Maintenance and Monitoring

GROUP 05 • PART F: REPOSITORY, TRACKER AND PROGRESS DOCUMENTATION

Cover field	Details
Prepared	6 November 2026; source-record consolidation
Group	05
Members	Daniel Cueto; Justine Philip Raguin; Aaron Imperial; Jenver Napenas; Emmanuel Manansala
Repository	https://github.com/grp05-dba/grp05_lostandfound
Activity 5 evidence	https://github.com/grp05-dba/grp05_lostandfound/tree/main/group-test-results/activity-5
Class tracker	https://docs.google.com/spreadsheets/d/15msFD5oWM6FzkEOyz3ayJP17TUayv2Qp/edit#gid=542211697
Document status	Editable draft. Activity 5 Log update, owner feedback and PDF submission not verified.

Evidence basis and scope

This document consolidates existing Group 05 Activity 5 Parts A–E, SQL files and captured command outputs from the main GitHub branch as reviewed on 6 November 2026. Activity 3 identifies the source applications; Activity 4 supplies the database accounts and least-privilege framework. The author of this consolidation did not connect to the local PostgreSQL instances or independently rerun tests. All observed results and limitations are attributed to the recorded project reports. No new maintenance or tracker edits were made to prepare this DOCX.

Repository overview — Group 05 application (verified from the repository README)

The group's own repository is the PUP-SPOT Campus Lost and Found system. Its README describes a PHP application with separate public and admin interfaces backed by PostgreSQL. This section documents the repository that hosts Group 05's work; it is distinct from the eight other-group databases evaluated in the Activity 5 monitoring sections below.

Repository and application facts

Item	Repository-documented details
Application	PUP-SPOT — Campus Lost and Found; browse found items, submit lost-item reports, request claims, and track submissions using reference codes.
Database	grp05_lostandfounddb (PostgreSQL).
Database scripts	sql/grp05_lostandfounddb_script.sql; sql/grp05_lostandfounddb_necessary_data_script.sql; sql/grp05_lostandfounddb_role_creation_sample_script.sql. Optional demo data: sql/grp05_lostandfounddb_demo_data_script.sql.
Application roles	lostandfound_app for the admin API and restricted lostandfound_public for the public API. The README says the role-creation script is run by a PostgreSQL administrator; application connections should not use the postgres superuser.
Local interfaces	Public server: php -S 127.0.0.1:8000 -t public. Admin server: php -S 127.0.0.1:8001 -t admin. The README notes that two local ports on one computer do not provide true network isolation.
Deployment/security note	The admin API is intended to be reachable only through a campus private network or equivalent restriction. The README states that the application does not include an admin username/password login; therefore, the admin server must not be exposed to the public internet.
Optional integrations	TextBee SMS and EmailJS email notifications are optional. Core database-backed workflows can run without them, but notification delivery requires valid private configuration.

Source and validation note: These application and setup details are taken from the public repository README on the main branch. They describe the documented design and setup instructions; they do not prove that the local database, PHP servers, role grants, or network controls were independently tested for this report. Repository README: https://github.com/grp05-dba/grp05_lostandfound

Activity 5 scope reminder: the assignment requires monitoring the other groups' Activity 3/4 projects, not treating Group 05's own PUP-SPOT database as one of those monitored targets. Group 05's repository is the location for this group's scripts, reports, and evidence.

Progress snapshot

Part F requirement	Evidence-based status
Step 1 — Own repository	Nine project folders (including Group 09 blocked prerequisite) contain reports or blocker notes; eight projects include executable SQL and supporting logs. Existing commits are not independently audited.
Step 2 — Activity 5 Log in tracker	Not verified; existing Part E summaries explicitly say tracker edits were not made.
Step 3 — PDF documentation	Working editable DOCX prepared. Export to PDF after review, then submit with the repository link.
External feedback	Owner-feedback drafts exist for evaluated projects; existing reports state feedback was not sent.
Coverage	Eight evaluated projects (02, 03, 04, 06, 07, 08, 10, 11); Group 09 blocked due to missing tracker repository link.

1. Projects monitored and accounts used

Group / project	Database / PostgreSQL	Account and role summary	Health / maintenance status
02 – Park It	grp_parkitdb / 18.6	park_it_auditor (readonly), park_it_dba (admin), park_it_app (runtime)	Completed maintenance; security issue unresolved
03 – OJT Management	ojt_managementdb / 18.6	grp03_ojt_auditor (readonly), grp03_ojt_dba (admin), grp03_ojt_app (runtime)	Completed maintenance; isolation issue unresolved
04 – BrewPOS	grp_brewposdb / 18.6	Existing runtime account; intended auditor/admin roles absent in inspected database	Maintenance blocked; role-plan mismatch
06 – HabalGo	grp00_habalgodb / 18.6	habalgo_readonly_user (readonly group), habalgo_app_user (runtime group)	Setup repaired; maintenance blocked
07 – CampusSafe	grp07_campussafedb / 18.6	campussafe_report (readonly), campussafe_app (runtime), campussafe_owner (maintenance owner)	Completed maintenance; connection-scope issue
08 – Campus Canteen	canteen_g05_activity4 / 18.3	canteen_admin (owner), canteen_app (runtime), Activity 4 reporting account	Selected-table maintenance completed
09 – Group 09 – Pending	Not established / N/A	No verified Activity 4 roles or project source	Blocked—missing source repository
10 – MESA	mesa_g05_activity4 / 18.3	rors_dba (owner), rors_app (runtime), rors_report (reporting)	Selected-table maintenance completed
11 – BoardTrack	boardtrack_g05_activity4 / 18.3	boardtrack_dba (owner), boardtrack_app (runtime), boardtrack_report (reporting)	Selected-table maintenance completed

Note: The databases above are separate local evaluation environments, not verified production deployments. The precise role memberships and SQL grants are in each project's Activity 4 files. Group 09 has no asserted database.

2. Health and activity evidence

Group 02 – Park It

Environment and inventory: grp_parkitdb; PostgreSQL 18.6; 5 tables; 5 sequences; 9 indexes.

Observation: Issues found: five tables empty; no current blocker. Broader PUBLIC CONNECT remains unresolved.

Evidence: https://github.com/grp05-dba/grp05_lostandfound/tree/main/group-test-results/activity-5/grp02-parkit/part_a.md; see project evidence subfolder for original health/activity outputs.

Group 03 – OJT Management

Environment and inventory: ojt_managementdb; PostgreSQL 18.6; 12 tables; 12 sequences; 24 indexes.

Observation: Issues found: database reachable, no blockers; access to unrelated databases remains possible.

Evidence: https://github.com/grp05-dba/grp05_lostandfound/tree/main/group-test-results/activity-5/grp03-Project-OJT-RMA-System/part_a.md; see project evidence subfolder for original health/activity outputs.

Group 04 – BrewPOS

Environment and inventory: grp_brewposdb; PostgreSQL 18.6; 5 tables; 4 sequences; 9 indexes.

Observation: Issues found: database reachable; 30 rows; missing planned Activity 4 maintenance/inspection identities.

Evidence: https://github.com/grp05-dba/grp05_lostandfound/tree/main/group-test-results/activity-5/grp04-brewpos/part_a.md; see project evidence subfolder for original health/activity outputs.

Group 06 – HabalGo

Environment and inventory: grp00_habalgodb; PostgreSQL 18.6; 8 tables; 7 sequences; 17 indexes.

Observation: Issues found: initially absent database/roles; locally provisioned; ten drivers, empty workflows; no blocker now.

Evidence: https://github.com/grp05-dba/grp05_lostandfound/tree/main/group-test-results/activity-5/grp06-habalgo/part_a.md; see project evidence subfolder for original health/activity outputs.

Group 07 – CampusSafe

Environment and inventory: grp07_campussafedb; PostgreSQL 18.6; 9 tables; 9 sequences; 24 indexes.

Observation: Issues found: locally provisioned setup, 129 seeded rows; integrity probes zero violations; unrelated postgres CONNECT remains possible.

Evidence: https://github.com/grp05-dba/grp05_lostandfound/tree/main/group-test-results/activity-5/grp07-campussafe/part_a.md; see project evidence subfolder for original health/activity outputs.

Group 08 – Campus Canteen

Environment and inventory: canteen_g05_activity4; PostgreSQL 18.3; 7 tables; 7 sequences; 11 indexes.

Observation: Reachable; 26 exact rows; one visible inspection session, no recorded blocker; Activity 4 ACLs preserved.

Evidence: https://github.com/grp05-dba/grp05_lostandfound/tree/main/group-test-results/activity-5/grp08-campus-canteen/part_a.md; see project evidence subfolder for original health/activity outputs.

Group 09 – Group 09 – Pending

Environment and inventory: Not established; PostgreSQL N/A; Not available.

Observation: Not reachable/not assessed: class Directory row lacked repository and application details when checked.

Evidence: https://github.com/grp05-dba/grp05_lostandfound/tree/main/group-test-results/activity-5/grp09-pending/README.md; see project evidence subfolder for original health/activity outputs.

Group 10 – MESA

Environment and inventory: mesa_g05_activity4; PostgreSQL 18.3; 8 tables; 7 sequences; 4 views; 19 indexes.

Observation: Reachable; 464 rows; one inspection session, no blocker; role ACLs preserved.

Evidence: https://github.com/grp05-dba/grp05_lostandfound/tree/main/group-test-results/activity-5/grp10-mesa/part_a.md; see project evidence subfolder for original health/activity outputs.

Group 11 – BoardTrack

Environment and inventory: boardtrack_g05_activity4; PostgreSQL 18.3; 8 tables; 7 sequences; 1 view; 16 indexes.

Observation: Reachable; 30 rows; one inspection session, no current blocker; ACLs preserved.

Evidence:
https://github.com/grp05-dba/grp05_lostandfound/tree/main/group-test-results/activity-5/grp11-boardtrack/part_a.md; see project evidence subfolder for original health/activity outputs.

3. Log findings and interpretations

Group 02: Service journal: 17 ERROR / 4 FATAL across test/history; SQL log-file functions denied, slow-query logging off. Interpretation and next step: distinguish expected negative privilege tests and scoped historical messages from ongoing application incidents; investigate repeatable unexplained errors through authorized log access. See part_b.md and evidence files.

Group 03: Scoped service journal: 24 project ERROR (22 expected prior negative tests, 2 current access probes); server-log SQL access denied. Interpretation and next step: distinguish expected negative privilege tests and scoped historical messages from ongoing application incidents; investigate repeatable unexplained errors through authorized log access. See part_b.md and evidence files.

Group 04: One historical administrative termination, 2 expected log-function denials and 2 maintenance-skip warnings; duration logging off. Interpretation and next step: distinguish expected negative privilege tests and scoped historical messages from ongoing application incidents; investigate repeatable unexplained errors through authorized log access. See part_b.md and evidence files.

Group 06: Two initial missing-role FATAL, two expected log-access ERROR, and two maintenance-skip WARNING; journal is scoped. Interpretation and next step: distinguish expected negative privilege tests and scoped historical messages from ongoing application incidents; investigate repeatable unexplained errors through authorized log access. See part_b.md and evidence files.

Group 07: One resolved initial missing-role failure and two expected log-function denials in available journal. Interpretation and next step: distinguish expected negative privilege tests and scoped historical messages from ongoing application incidents; investigate repeatable unexplained errors through authorized log access. See part_b.md and evidence files.

Group 08: Scoped retention: 66 ERROR / 9 FATAL; expected testing categories, no verified production incident; duration logging unavailable. Interpretation and next step: distinguish expected negative privilege tests and scoped historical messages from ongoing application incidents; investigate repeatable unexplained errors through authorized log access. See part_b.md and evidence files.

Group 09: Not checked: no verified database or log source. Interpretation and next step: distinguish expected negative privilege tests and scoped historical messages from ongoing application incidents; investigate repeatable unexplained errors through authorized log access. No server-log data was available because the project is blocked.

Group 10: 64 ERROR / 9 FATAL; expected testing categories, no unexplained production incident; no duration logging. Interpretation and next step: distinguish expected negative privilege tests and scoped historical messages from ongoing application incidents; investigate repeatable unexplained errors through authorized log access. See part_b.md and evidence files.

Group 11: 66 ERROR / 9 FATAL; test-related categories, no verified production incident; duration logging unavailable. Interpretation and next step: distinguish expected negative privilege tests and scoped historical messages from ongoing application incidents; investigate repeatable unexplained errors through authorized log access. See part_b.md and evidence files.

Scope caution: available service journals and dedicated log windows are incomplete incident histories. No failed-password or slow-query incident should be inferred unless recorded by a specific source.

4. Maintenance performed and verification

Project	Actual action / outcome
02 – Park It	ANALYZE and ordinary VACUUM on all five tables succeeded under project DBA; 2 dead vehicle tuples removed. Fresh counters confirmed.
03 – OJT Management	12 ANALYZE and 12 ordinary VACUUM operations completed under owner/admin; 18 dead tuples removed; seeded live rows preserved.

Project	Actual action / outcome
04 – BrewPOS	ANALYZE and ordinary VACUUM attempted on menu_item, but both SKIPPED for insufficient MAINTAIN permission. Not completed.
06 – HabalGo	ANALYZE/VACUUM attempted on driver with runtime login; both SKIPPED for insufficient maintenance privileges. Not completed.
07 – CampusSafe	Nine ANALYZE + nine ordinary VACUUM commands completed under non-superuser owner; 129 rows preserved; fresh counters checked.
08 – Campus Canteen	ANALYZE and ordinary VACUUM/ANALYZE on customer_order completed under owner; counters verified, counts and ACLs unchanged.
09 – Group 09 – Pending	Not performed: no database to authorize or maintain.
10 – MESA	ANALYZE + ordinary VACUUM/ANALYZE on menu_item completed; fresh counters, counts/sequences/ACLs unchanged.
11 – BoardTrack	ANALYZE six business tables, ordinary VACUUM/ANALYZE on lease completed; counters and exact counts verified.

The source reports distinguish completed ordinary VACUUM/ANALYZE from skipped commands. VACUUM FULL, speculative index rebuilds, and broad grants to normal runtime users are not presented as performed. Evidence: each project's part_c.md plus maintenance and follow-up verification outputs.

5. Query performance analysis

Project	Recorded plan and interpretation
02 – Park It	Source monitoring SELECT: sequential scan, nested loops and sort; 0 rows, ~0.104 ms (empty sample).
03 – OJT Management	Student requirements SELECT: sequential scans, hash left join and sort; 3 rows, ~0.110 ms.
04 – BrewPOS	Daily-sales SQL translation: sequential scans, hash join, sort, aggregate; ~0.123 ms on tiny sample.
06 – HabalGo	Driver dropdown SELECT: scan and sort; estimate 190 vs actual 10; ~0.105 ms; initial statistics require refresh.
07 – CampusSafe	Category summary SELECT: scans, hash join, hash aggregate and sort; 6 groups, ~0.185 ms.
08 – Campus Canteen	Source SELECT: root Aggregate; estimate/actual 2/2, sampled ~0.446 ms.
09 – Group 09 – Pending	Not performed: no project/query available.
10 – MESA	Menu SELECT: root Sort, estimated/actual 48/48, sampled ~0.472 ms.
11 – BoardTrack	Lease SELECT: root Sort; estimated/actual 4/4, sampled ~0.882 ms.

These are small local datasets; recorded millisecond timings do not establish production performance. Existing reports generally recommend monitoring larger representative workloads before new indexes or query rewrites. Source execution plans and exact statements remain in part_d.md and linked evidence files.

6. Prioritized recommendations

Group	Priority	Recommended action
02	Immediate action	Immediate: review cross-database PUBLIC CONNECT with authorized administrator. Follow-up: seed representative data, verify source provenance and reassess plans.
03	Immediate action	Immediate: correct inappropriate cross-database CONNECT. Follow-up: automated audit logging, reproducible migrations and representative workloads.
04	Immediate action	Immediate prerequisite: restore/reconcile Activity 4 role plan, then complete authorized ANALYZE/VACUUM. Follow-up: PUBLIC CONNECT and statistics.
06	Follow-up action	Follow-up: approve a scoped maintenance role and rerun work, verify stats; review

Group	Priority	Recommended action
		supplied password literals, PUBLIC CONNECT and runtime config.
07	Follow-up action	Follow-up: review unrelated-database CONNECT, upstream owner/role installer, future-table default privileges and representative workload.
08	Follow-up action	Follow-up: monitor autovacuum/row growth, correlate abandoned sessions and benchmark order/vendor predicates with realistic volume.
09	Immediate prerequisite	Immediate prerequisite: obtain Group 09 repository/project details from class tracker, then complete Activity 4 setup before Activity 5.
10	Follow-up action	Follow-up: monitor growth/autovacuum; add scoped performance logging only if needed; benchmark indexes at scale.
11	Follow-up action	Follow-up: review abandoned sessions, observe growth/autovacuum and benchmark repeated lease aggregates at scale.

7. Limitations, permissions and Activity 4 relationship

Group 04 and Group 06 maintenance attempts were skipped by PostgreSQL: current project credentials did not have sufficient MAINTAIN/ownership rights. The reports correctly refrain from escalating to a superuser merely to make commands succeed. Group 02 and Group 03 retained overly broad cross-database CONNECT access; Group 07 also noted unrelated database CONNECT visibility. Authorized administrators should review legitimate clients and cluster-wide PUBLIC defaults before changing them. Some monitoring views and server logs are inaccessible to the ordinary accounts. These are documented privilege or observability limits, not evidence that the relevant data were reviewed.

Where local test setup had to be adapted or created anew (notably Groups 06 and 07), the results do not prove that the owning group's Activity 4 deployment was complete or configured identically. Testing generally uses small seed data and limited session snapshots, so concurrency, realistic workload behavior and production incident history remain unknown.

8. Feedback to owning groups

The project findings provide feedback drafts, but the reviewed source summaries explicitly say they have not been delivered. Enter the sent date only after the owning group has actually been contacted.

Group	Feedback summary	Sent? / date
02	Immediate: review cross-database PUBLIC CONNECT with authorized administrator. Follow-up: seed representative data, verify source provenance and reassess plans.	NO (as reported) / Date: _____
03	Immediate: correct inappropriate cross-database CONNECT. Follow-up: automated audit logging, reproducible migrations and representative workloads.	NO (as reported) / Date: _____
04	Immediate prerequisite: restore/reconcile Activity 4 role plan, then complete authorized ANALYZE/VACUUM. Follow-up: PUBLIC CONNECT and statistics.	NO (as reported) / Date: _____
06	Follow-up: approve a scoped maintenance role and rerun work, verify stats; review supplied password literals, PUBLIC CONNECT and runtime config.	NO (as reported) / Date: _____
07	Follow-up: review unrelated-database CONNECT, upstream owner/role installer, future-table default privileges and representative workload.	NO (as reported) / Date: _____
08	Follow-up: monitor autovacuum/row growth, correlate abandoned sessions and benchmark order/vendor predicates with realistic volume.	NO (as reported) / Date: _____

Group	Feedback summary	Sent? / date
10	Follow-up: monitor growth/autovacuum; add scoped performance logging only if needed; benchmark indexes at scale.	NO (as reported) / Date: _____
11	Follow-up: review abandoned sessions, observe growth/autovacuum and benchmark repeated lease aggregates at scale.	NO (as reported) / Date: _____
09	Obtain repository/owner contact from tracker before any evaluation.	Not applicable / pending

9. AI usage declaration and validation

Recorded source reports disclose OpenAI Codex assistance in reading instructions/source and Activity 4 plans, preparing SQL, executing or interpreting local checks, reviewing service log excerpts, and drafting Parts A–E. Their validation evidence includes distinct authenticated sessions, SQL output, role/ACL inspection, maintenance warnings or completion output, fresh counters, unchanged exact row counts, and source-derived EXPLAIN plans. ChatGPT additionally consolidated the existing reports into this editable Word file. That consolidation did not independently execute database queries or validate live server state. Human confirmation remains required before submission; no secrets should be reproduced in screenshots, logs, documents, or tracker rows.

10. Part F submission checklist (editable)

Required action	Current status / owner update
Confirm all tracker-listed groups	Eight evaluated; Group 09 blocked pending repository link. Recheck tracker. [_____]
Review database roles against Activity 4	Source files document roles; exceptions and missing roles disclosed. [_____]
Confirm health and log evidence	Present for eight groups with limitations. [_____]
Confirm successful maintenance per project	Group 04 and 06 incomplete due to permissions; reattempt only with authorized role. [_____]
Confirm query plans and recommendations	Plans recorded for eight groups; Group 09 blocked. [_____]
Update Activity 5 Log tab	Not verified / reportedly not done. [_____]
Send feedback to owning groups	Not sent according to source reports. [_____]
Review group member names and tracker link	Carried forward from the user-updated Activity 4 draft; reconfirm before submission. [_____]
Secrets screening	Perform final manual review of linked evidence/screenshots. [_____]
Export and submit PDF and repository link	DOCX draft only; final submission not recorded. [_____]

Document source links

Activity 5 instructions: user-provided COMP018_Activity_5(1).docx, Part F (pages 4–6). Project evidence root: https://github.com/grp05-dba/grp05_lostandfound/tree/main/group-test-results/activity-5. Each project folder contains part_a.md–part_e.md (Group 09: README.md blocker), scripts and captured evidence. Activity 3/4 context is retained in the same Group 05 repository.